



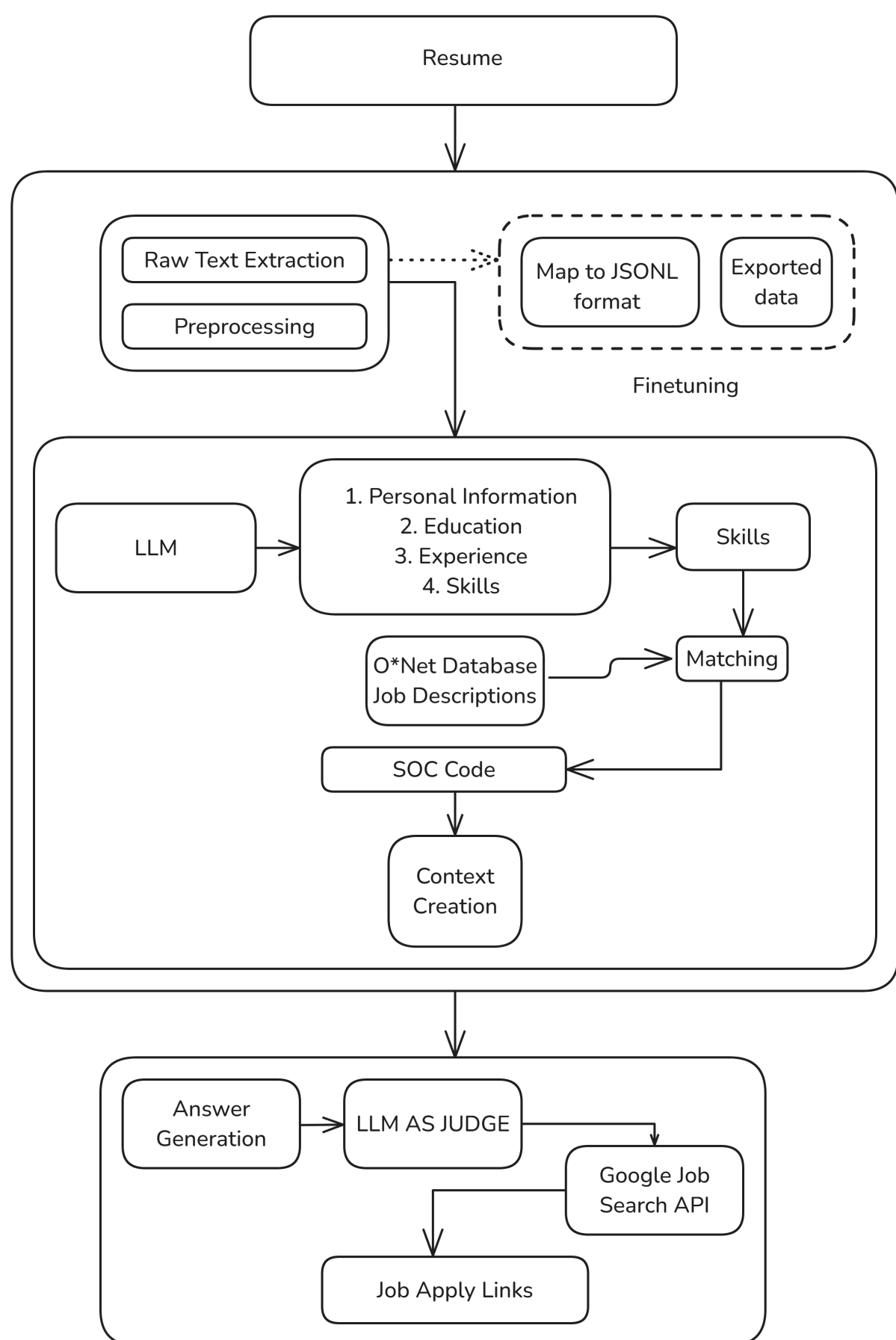
Enhancing Career Guidance for Students through Retrieval Augmented Generation

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Introduction

- This research explores the use of **Retrieval-Augmented Generation (RAG)** for personalized career guidance by comparing the performance of three large language models (LLMs): **Gemma 2 9B**, **Gemma 2 2B** and **Mixtral 8x22B**.
- Existing career guidance chatbots provide only general recommendations, but this study leverages **FAISS retrieval** and the **O*NET job description dataset** to generate tailored career advice from user resumes.
- Gemma 2 9B** and **Gemma 2 2B** developed by Google DeepMind, is a *9-billion-parameter* model optimized for efficient performance. **Mixtral 8x22B**, a *sparse Mixture-of-Experts (SMoE)* model from Mistral AI, utilizes *39B active parameters*, making it faster than many dense 70B models.
- FAISS (Facebook AI Similarity Search)** is used for efficient similarity search and clustering of dense vectors.
- The **O*NET dataset** is embedded into dense vectors using a pretrained model, allowing FAISS to match user skills with the most relevant job descriptions.
- This study is among the first to integrate these advanced LLMs with FAISS for career guidance.
- By retrieving the best job matches based on user resumes and generating tailored recommendations, these **RAG-powered systems** provide *more precise and personalized* career advice than traditional models.
- The research evaluates their performance in delivering specialized guidance, offering insights into their effectiveness in real-world applications.

Architecture Diagram



Career Guidance Generation Module

- Personalized career recommendations are produced by the **Mistral-based Career Guidance Generation system using a structured prompt**.
- In order to give pertinent context, it uses a Retrieval-Augmented Generation(RAG) technique, in which the top-K documents are obtained via FAISS and O*NET job titles/description.

Resume Parser

- The Gemma2 2B model was effectively fine-tuned using Parameter-Efficient Fine-Tuning (PEFT) and Low-Rank Adaptation (LoRA). LoRA, a low-rank decomposition technique, speeds up fine-tuning without sacrificing model performance by drastically lowering the number of trainable parameters.
- Large pretrained models can be effectively adapted with this method's low computational overhead. The model produces structured data in JSONL format, which facilitates easy integration with other system elements.
- Data Collection:** Database of job descriptions and resumes is collected.
- Pre-processing:** JSONL files containing labeled annotations were created from the data
- Training:** To increase accuracy and make sure the model complies with domain-specific standards. We have used UNSLOTH.AI platform to optimize.

Data Collection Module

- The resume parsing process was automated using **Gemma 2 2B**, which accurately extracted important information such as name, email, phone number, skills, education, and experience.
- Text extraction, NLP-based preprocessing, model-based parsing, and Named Entity Recognition (NER)-generated structured output are all part of the methodology. The results demonstrate great information extraction precision and 80% quicker processing.



Figure 1. Training vs loss

Performance Metrics Across Models

Metric	Gemma 2 2B	Gemma 2 9B	Mistral
BLEU	0.07	0.08	0.17
ROUGE	0.21	0.24	0.25
BERT	0.85	0.85	0.87
SMS	0.77	0.79	0.87
BLEURT	-0.25	-0.17	-0.02

Table 1. Performance Metrics Comparison Across Models

The observed results indicate that Mistral outperforms both the Gemma models. The reference text used as ground truth for comparing performances of each of the model was generated by Deepseek-R1 14B. All three models display a very high score of BERT and SMS which indicates a high good semantic similarity. The BLEU and ROUGE scores indicates scope for improvement in fluency and recall. The negative BLEURT score shows that these models are still far behind to produce a human-like, realistic output like the Deepseek-R1 14B.

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